

Robot Autonomy - HW1 - Control

Submission guidelines: Submission of this assignment is in pairs. You must submit a single zip file.

Project setup guide:

1. Install Python

Go to the official Python website: <https://www.python.org/downloads/>.

Download **Python 3.10**

During installation, **check the box “Add Python to PATH”**.

2. Install an Editor

You'll need a code editor or IDE. Recommended: PyCharm

3. Get the Project Files

Place the following files in a single project folder

4. Install Required Packages

This project depends on **numpy** and **matplotlib**. Install them using the terminal command:

```
pip install numpy matplotlib
```

5. Run the Simulations

(a) PID Wall Following

```
python PID_wall_following.py
```

Q1: PID

You are required to design and tune a PID controller that controls the steering angle of a car to drive it along a wall (keeps 1m from the wall).

1. Implement PID class in PID_wall_following.py.
2. Set the parameters $k_p=0.3$, $k_d=0$, $k_i=0$. Report the error graph and average error value. Repeat for $k_p=1.0$, $k_d=0$, $k_i=0$. Explain the differences.
3. Set the parameters $k_p=0.3$, $k_d=0$, $k_i=0$. Report the error graph and average error value. Repeat for $k_p=0.3$, $k_d=0.2$, $k_i=0$. Explain the differences.
4. Now we add constant error to steering execution. Set $\text{STEERING_ERROR_CONST} = 1.0$. and $k_p=1.5$, $k_d=0.05$, $k_i=0.0$. Report the error graph and average error value. Repeat for $k_p=1.0$, $k_d=0.05$, $k_i=0.5$. Explain the differences.
5. To account for noise in the real world we assume inaccuracy of localization and steering execution, Set $\text{STEERING_ERROR_CONST} = 0$, $\text{STEERING_ERROR_PERCENTAGE} = 0.3$, and $\text{LOCALIZATION_ERROR_METER} = 0.5$. Report the error graph and average error value. Tune k_p , k_d , k_i as you wish and provide 3 examples. Report the error graph. Suggest a method to tune the parameters k_p , k_d , k_i .

Q2: Pure Pursuit

You are required to implement a pure pursuit controller to track a given path.

1. Implement the PurePursuit_controller class in PurePursuit.py. You may change the class and functions as you wish. Do not change other parts of the code.
2. Set lookahead distance $L_f=0.3$ and report tracking graph and error graph.
3. Set lookahead distance $L_f=1.0$ and report tracking graph and error graph. Explain the differences with the previous results.
4. Suggest a method to tune the lookahead distance parameter, test and explain your method, and provide a tracking and error graph.

